Chapter 16

Acid-Base Equilibria

1. Which one of the following is a BrØnsted-Lowry acid? ____E____

A) $(CH_3)_3NH^+$
B) CH_3COOH
C) HF
D) HNO_2
E) all of the above

Brønsted–Lowry (Acid-base proton theory):

1) Acid is a substance that donates a proton to another substance;
2) Base is a substance that accepts a proton from another substance;

2. Which one of the following statements regarding K_w is false? B

A) pK_w is 14.00 at 25 ^{o}C.
B) The value of K_w is always 1.0×10^{-14}.
C) K_w changes with temperature.
D) K_w is known as the ion product of water.

Ion-product constant for water:

$K_w = [H_3O^+][OH^-] = [H^+][OH^-] = 1.0\times10^{-14}$ （T=25°C）

3. Which one of the following is the weakest acid? ____D____

A) HF ($K_a = 6.8 \times 10^{-4}$)
B) HClO ($K_a = 3.0 \times 10^{-8}$)
C) HNO_2 ($K_a = 4.5 \times 10^{-4}$)
D) HCN ($K_a = 4.9 \times 10^{-10}$)
E) CH_3COOH ($K_a = 1.8 \times 10^{-5}$)

2) Acid-dissociation constant: $\mathbf{K_a}=[H^+][A^-]/[HA]$;
3) The greater the value of K_a, the stronger is the acid;

4. Which of the following acids will be the strongest? ____A____

A) H_2SO_4

B) HSO_4^- แตกตัวอีกรอบ K_a น้อย

C) H_2SO_3

D) H_2SeO_4

E) HSO_3^-

5. The Ka of hypochlorous acid (HClO) is 3.0×10^{-8} at 25 °C. What is the percent ionization (%) of hypochlorous acid in a 0.015 M aqueous solution of HClO at 25 °C? ___D___

$HClO \longrightarrow ClO^- + H_3O^+$

$\frac{x^2}{0.015} = 3 \times 10^{-8}$

$x = 2.12 \times 10^{-5}$

A) 4.5×10^{-8}
B) 14
C) 2.1×10^{-5}
D) 0.14
E) 1.4×10^{-3}

6. A^- is a weak base. Which equilibrium corresponds to the equilibrium constant Ka for HA? ____C____

A) $HA\,(aq) + H_2O\,(l) \rightleftharpoons H_2A^+\,(aq) + OH^-\,(aq)$

B) $A^-\,(aq) + H_3O^+\,(aq) \rightleftharpoons HA\,(aq) + H_2O\,(l)$

C) $HA\,(aq) + H_2O\,(l) \rightleftharpoons H_3O^+\,(aq) + A^-\,(aq)$

D) $A^-\,(aq) + H_2O\,(l) \rightleftharpoons HA\,(aq) + OH^-\,(aq)$

E) $A^-\,(aq) + OH^-\,(aq) \rightleftharpoons HOA_2^-\,(aq)$

react with water to form $OH^-(aq)$;

Weak acids: $HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$

1) Weak acids only partially ionized to ions in aqueous solution;
2) Acid-dissociation constant: $K_a=[H^+][A^-]/[HA]$; 酸常数

7. Using the data in the table, which of the conjugate acids below is the strongest acid? ____C____

A) HClO
B) HCO_3^-
C) H_2S K_a เยอะสุด (K_b น้อยสุด)
D) $NH_3CH_3^+$
E) H_2S and HClO

Base	K_b
ClO^-	3.3×10^{-7}
CO_3^{-2}	1.8×10^{-4}
HS^-	1.8×10^{-7}
NH_2CH_3	4.4×10^{-4}

8. Which of the following aqueous solutions has the highest $[OH^-]$?

____D____

pH เยอะสุด

A) a solution with a pH of 3.0

B) a 1×10^{-4} M solution of HNO_3 pH = 4

C) a solution with a pOH of 12.0 pH = 2

D) pure water pH = 7

E) a 1×10^{-3} M solution of NH_4Cl

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10. Of the compounds below, a 0.1 M aqueous solution of ____A____ will have the highest pH. K_a น้อยสุด = pH เยอะสุด

A) KCN, K_a of HCN = 4.0×10^{-10}

B) NH_4NO_3, K_b of NH_3 = 1.8×10^{-5} $K_a = 5.6 \times 10^{-10}$

C) NaOAc, K_a of HOAc = 1.8×10^{-5}

D) NaClO, K_a of HClO = 3.2×10^{-8}

E) NaHS, K_b of HS^- = 1.8×10^{-7}

$K_a = 5.6 \times 10^{-8}$